

HRUTHIK JAYARAMA

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MS ECE student at Portland State specializing in design verification. Hands-on with SystemVerilog, UVM and Assertion based testbench development, with strong understanding in computer architecture fundamentals.

PROJECTS

Tomasulo's Algorithm – Cycle-Accurate Out-of-Order Execution Simulator October 2025

- Implemented a cycle-accurate Tomasulo simulator with reservation stations, register renaming, and CDB to model dynamic out-of-order instruction scheduling and instruction latencies.
- Gained deep understanding of out-of-order execution, how reservation stations, register renaming, and CDB work together to eliminate WAR/WAW hazards.

Asynchronous FIFO – UVM Functional Verification April 2025

- Designed a parameterized asynchronous FIFO with configurable depth and status flags (full, empty); built a fully reusable UVM environment with constraint-random stimulus generation.
- Understood clock domain crossing challenges and how to verify them, built a complete UVM environment with constraint-random stimulus and functional coverage.

MIPS-Lite 5-Stage Pipeline Simulator – Functional and Timing Simulation April 2025

- Built a functional and timing simulator for a MIPS-lite ISA with a 5-stage pipeline (IF, ID, EX, MEM, WB), RAW hazard detection, branch misprediction handling, and configurable data forwarding.
- Gained understanding of pipeline hazards, data forwarding paths, and how architectural decisions directly impact IPC, verified through trace-level analysis.

2x2 Router / Arbiter – Assertion-Based Formal Verification October 2025

- Verified a parameterized 2x2 router using VC Formal and JasperGold; wrote SVA assertions, assumptions, and cover properties via bind files to keep the testbench decoupled from the DUT.
- Understood the full FPV debug cycle, writing bind-file environments, reading counterexample traces to find RTL bugs, and iterating to formal closure.

Cache Controller – SystemVerilog January 2025

- Designed a 4-core split L1 cache (4-way instruction, 8-way data, LRU) with a shared inclusive L2; implemented the full MESI coherence protocol for multi-core cache state management.
- Built strong intuition for cache coherence. implemented and verified all MESI state transitions in a 4-core system, understanding LRU replacement and hit/miss behavior.

SKILLS

Hardware Description / Verification Languages: SystemVerilog, Verilog, UVM, Python

Verification Methodologies: Functional Verification, Assertion-Based Verification (ABV), Constraint-Random Stimulus, Coverage-Driven Verification (CDV), Gate-Level Simulation, Pre-Silicon Verification

UVM Components: UVM Environment, Coverage Groups, Sequences, Virtual Sequences

EDA Tools: Siemens QuestaSim, Synopsys VC Formal, Cadence JasperGold, Xilinx Simulator, Pspice

Verification Concepts: UVM & OOP Methodology, Assertions operators and structure

Protocols & Interfaces: UART, I2C, SPI, MESI, MSI

Computer Architecture: Cache Coherence, Virtual Memory, Pipelining, Out-of-Order Execution, Branch Prediction, MIPS ISA

Operating Systems: Linux, Windows

EDUCATION

Portland State University, Portland, OR January 2025 – December 2026

M.S., Electrical and Computer Engineering GPA: 3.75/4.0

Sai Vidya Institute of Technology (VTU University) December 2020 – June 2024

B.E., Electronics and Communication Engineering

CERTIFICATIONS

- Essential SystemVerilog for UVM – Cadence
- System Verilog for Design and verification – Cadence
- Design for Test Fundamentals — Cadence